

Demonstration of Proposed Features

Date	15 July 2026
Team ID	SWUID2026217449
Project Name	Rising Waters - Flood Prediction ML
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Flood Risk Prediction (High/Low)	Predicts flood risk based on 5 weather indicators	Implemented	Yes	Core feature, working as expected
2	Multi-Model Comparison	Trains and compares Decision Tree, Random Forest, KNN, XGBoost	Implemented	Yes	Random Forest selected as best model
3	Data Preprocessing & Scaling	Cleans and scales weather data before training	Implemented	Yes	StandardAero used
4	Web Interface (Flask + HTML)	User-friendly form to input weather data and view result	Implemented	Yes	Built with Flask, index and result pages
5	Model Deployment (.pkl integration)	Trained model and scaler saved and loaded for real-time prediction	Implemented	Yes	Uses model and scalar
6	Scenario-based Testing	Validated predictions using high-risk and low-risk scenarios	Implemented	Yes	Both outcomes verified successfully

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%